St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank {Revised Curriculum (LOCF) - 2020-22 Batch }

Second Year- Third Semester

Course Title (Paper Title):

INSTRUMENTAL METHODS OF ANALYSIS.QP Code: 85321

Unit	SI. No	Questions	Marks
3	1.	List out the various techniques involved in thermal methods of analysis.	2
3	2.	Mention the various techniques involved in thermal methods of analysis.	2
3	3.	Write the various techniques involved in thermal methods of analysis.	2
3	4.	Give the types and principles of thermogravimetric analysis.	2
3	5.	Write the types and principles of thermogravimetric analysis.	2
3	6.	Explain the types and principles of thermogravimetric analysis.	2
3	7.	What are plateaus and inflections?	2
3	8.	Give the principles of thermogravimetric analysis.	2
3	9.	Write the principles of thermogravimetric analysis.	2
3	10.	Explain the principles of thermogravimetric analysis.	2
3	11.	Give the applications of TGA.	2
3	12.	Write the applications of TGA.	2
3	13.	Mention the applications of TGA.	2
3	14.	Write the theory of DTA.	2
3	15.	Give the theory of DTA	2
3	16.	Give the difference between DTA and DSC.	2
3	17.	Write the difference between DTA and DSC.	2
3	18.	Differentiate DTA and DSC.	2
3	19.	Give the differences between TGA and DTA.	2
3	20.	Differentiate TGA and DTA.	2
3	21.	Write the differences between TGA and DTA.	2
3	22.	Give the general principles of DTA.	2
3	23.	Write the general principles of DTA.	2
3	24.	Explain the general principles of DTA.	2
3	25.	Mention the applications of DTA.	2
3	26.	Outline the applications of differential thermal analysis.	2
3	27.	Write the applications of DTA.	2
3	28.	List out the applications of differential thermal analysis.	2
3	29.	Give the applications of DTA.	2
3	30.	Give the basic principles of DSC.	2
3	31.	Write the basic principles of DSC.	2
3	32.	Explain the basic principles of DSC.	2
3	33.	Mention the applications of DSC.	2

3	34.	Outline the applications of DSC.	2
3	35.	Write the applications of DSC.	2
3	36.	List out the applications of DSC.	2
3	37.	Give the applications of DSC.	2
3	38.	What are enthalpimetric titrations? Give their principle.	2
3	39.	Give the principle and theory of thermometric titration.	2
3	40.	What are thermometric titrations? Give their principle.	2
3	41.	Write the principle and theory of thermometric titration.	2
1	42.	What are the differences between atomic absorption and flame emission spectroscopy?	2
1	43.	What are the advantages of atomic absorption spectroscopy over flame emission spectroscopy?	2
1	44.	How will you determine metallic elements in the food industry?	2
1	45.	Define detection limit and sensitivity of an atomic absorption spectroscopy	2
1	46.	How will you determine calcium, magnesium, sodium, and potassium in blood serum?	2
1	47.	Compare the Nephelometer, turbidimeter, and spectrophotometer with reference to the sample of analysis	2
1	48.	Compare the Nephelometer, turbidimeter, and spectrophotometer instruments as to the geometrical arrangements of the samples and detection.	2
1	49.	Distinguish fluorescence and phosphorescence in terms of the mechanism by which an excited molecule is deactivated.	2
1	50.	What are the two basic differences in the construction of a fluorimeter as compared to an absorption spectrophotometer?	2
1	51.	Define singlet and triplet states	2
1	52.	Compare fluorimetry and phosphorimetry with absorption methods.	2
1	53.	What is the main criterion for deciding whether turbidimetry or Nephelometry should be used in the analysis of a medium containing suspended particles?	2
1	54.	Why is the atomization process carried out in AAS? How the AAS differs from molecular absorption spectrometry?	2
1	55.	In neutral or alkaline solution aniline exhibits fluorescence in the visible region and this visible fluorescence disappears if the aniline is acidified. Give the reasons	2
1	56.	Draw the schematic diagram of turbidometer and Nephelometer, and label the components.	2
1	57.	Write the principle of turbidimetry	2
1	58.	Define the term atomization. Why is it regarded as a critical step in the atomic spectrophotometric method of analysis?	2
1	59.	Why are the most fluorescence instruments double beam in design?	2
1	60.	What is an internal standard and why is it used?	2
1	61.	Distinguish singlet excited state from triplet excited state	2
1	62.	Suggest any two methods to increase the sensitivity of determination in AAS.	2

1	63.	Give the expression for estimation of the relative population of the ground state and excited state population at a given flame temperature	2
1	64.	Compare the fluorescence intensity of aniline (neutral medium) and benzoic acid	2
1	65.	How atomic absorption spectroscopy is different from atomic emission spectroscopy in principle and instrumentation?	2
1	66.	What are the advantages of flame atomisers in AAS?	2
1	67.	How does the absorption spectrum distinguish from the emission spectrum for sodium?	2
1	68.	Why flame emission spectrometry is often as sensitive as atomic absorption spectrometry?	2
1	69.	Define the process of intersystem crossing	2
1	70.	Name any two oxidants and fuels used in FES	2
1	71.	How is atomic absorption spectrophotometry different from flame emission spectrometry?	2
1	72.	What are events occur in rapid succession when a solution of NaCl is sprayed into the flame?	2
1	73.	Evaluate the turbidimetric coefficient like molar absorptivity if a sample was analyzed by turbidimetry with transmittance as 0.368 cell path length of 1 cm and concentration was 112 mg/L?	2
1	74.	Calculate the turbidence value if the intensity of incident radiation was 100 and the relative intensity of transmitted radiation was 45.2?	2
1	75.	Find out the concentration of sulphate if turbidimetric analysis of sulphate was carried out at 355 mm with an optical path of 1 cm with the turbulence (i.e., absorption) 0.121, with turbidity coefficient was 1.73×10^3 L/mol/cm at 355 mm?	2
1	76.	Calculate the amount of sulphate present in the unknown sample if a standard solution containing 7.59 mg/250 ml had a reading of 5.0%. The unknown solution had a reading of 4.26%?	2
1	77.	An unknown compound was measured by fluorimetry with a standard (1µg/ml) having a fluorescence intensity of 73. If the unknown solution showed an intensity of 62 what was its concentration?	2
1	78.	A beryllium-morine complex (Be = 200 mg/ml) had an intensity of 78.8 with two units for the blank solution. What was the concentration of unknown it had an intensity of 63.5 units?	2
1	79.	Aluminium with oxine showed strong fluorescence. A standard solution of 5 mg/ml of aluminium showed an intensity of 0.450, what was the concentration of aluminium in an unknown sample if its intensity was 0.990.	2
1	80.	A 0.150 mg of selenium showed fluorescence of 0.250 with 2,2-diaminobenzidine. If 50 mg of gallium was added fluorescence rose to 0.750. Determine % gallium in selenium if blank showed fluorescence of 0.05 units.	2
1	81.	What is the quantum efficiency if 4.9×10^{17} photons were absorbed during the excitation and 2.9×10^{17} photons were emitted in fluorescence?	2
1	82.	Absorbed radiation was reemitted with no change in the intensity such that the typical emission at the longer wavelength was named by definite name?	2

		What was the name and what was the kind of shift noted?	
1	83.	What is the wavelength of the spectral line resulting in the transition from 4p back to 4s ground state-provided if the excitation potential for 4p level is 2.95 eV?	2
1	84.	What is the wavelength for transition if for flame excited lithium atom; two transitions occur at 3p to 2s and 4d to 2p with energy difference of 3.85 to 0 eV and 4.60 to 1.90 eV, respectively?	2
1	85.	What is the wavelength of the resonance line of the sodium atom if the excitation energy of the resonance level was 2.10 eV?	2
1	86.	The excitation potential for the 4p level in calcium was 2.95 eV. Find out the wavelength of the spectral line resulting from the transition from 4p back to 4s ground state?	2
1	87.	**Calculate the two-wavelength corresponding to two electronic transitions as lithium shows two transitions namely, 3p to 2s (3.8 to 0.0 eV) and 4d to 2p (4.60 to 1.90 eV)?**	2
2	88.	Explain the terms a) charging current and b) migration current	2
2	89.	Point out the differences between the experimental conditions used in ordinary electrogravimetry and polarography	2
2	90.	What are the advantages of the DME over a solid microelectrode? Compare the usefulness of the DME as a cathode and as an anode	2
2	91.	Explain what is meant by a completely polarised electrode. Can the reaction of such an electrode be reversible?	2
2	92.	Give two reasons for using a supporting electrolyte in voltammetry	2
2	93.	Explain why the total cathode potential is the only term affected by a change in the applied voltage under polarographic conditions	2
2	94.	Give three reasons for using dropping mercury electrodes in polarography	2
2	95.	Define half-wave potential, diffusion current, depolarizer DME, residual current, and voltammetry	2
2	96.	Which of the following parameters in polarography is best used as a qualitative tool: Half wave potential, Migration current, diffusion current, and decomposition potential	2
2	97.	Discuss the factors that may contribute to the observed polarographic limiting current of a single ion in addition to the diffusion current. How may all the undesirable factors be eliminated?	2
2	98.	Polarographic methods resemble potentiometric titration curves. What are the advantages of polarography over potentiometry	2
2	99.	What is the underlying principle in amperometric titrations	2
2	100.	What are the advantages and disadvantages of amperometric titrations over other methods	2
2	101.	What are the advantages of rotating platinum microelectrode over dropping mercury electrode	2
2	102.	What is the principle of electrogravimetry?	2
2	103.	Describe the process of electrodeposition.	2
2	104.	Write a short note on a) the counter or back potential b) decomposition potential c) overvoltage	2
2	105.	Briefly discuss the applications of electrogravimetry	2

2	106.	What is a glass electrode? Explain its principle	2
2	107.	Briefly discuss the cause acidic error of the glass electrode	2
2	108.	Explain how the glass electrode functions as an indicator electrode for hydrogen ions	2
2	109.	Discuss the pH limitations of ion selective electrodes	2
2	110.	Explain how the mercury electrode serves as an indicator electrode for various metal ions.	2
2	111.	Define the first and second laws of Faraday's law of electrolysis.	2
2	112.	Define three types of current developed in the voltammetric procedure	2
2	113.	Explain indicator electrodes of the second kind with an example	2
2	114.	List the advantages and disadvantages of coulometric titration as compared to conventional volumetric titration	2
2	115.	Define three different modes of current in voltammetry. Draw the graphical plots of diffusion-controlled and limiting current in voltammetry	2
2	116.	Define three mechanisms responsible for the transport of dissolved species to and from the electrode surface.	2
2	117.	Define kinetic polarization and briefly explain the precautions that should be taken for controlling it.	2
2	118.	Explain reaction and principle of acid-base titration by coulometric methods of analysis involving generator electrode	2
2	119.	Give the classification of electrochemical methods of analysis and mention the quantity measured in each of them.	2
2	120.	Define the terms: ohmic potential and decomposition potential in electrogravimetry.	2
2	121.	Define anodic current and cathodic current	2
2	122.	What is liquid junction potential?	2
2	123.	Write the difference between galvanostat and potentiostat	2
2	124.	Write the classification of the voltammetric technique.	2
2	125.	Bring out the difference between an electrolytic cell and a galvanic cell	2
2	126.	Is the Electrogravimetric method absolute or reference? Justify your answer	2
2	127.	Sketch the waveform excitation signal for cyclic voltammetry	2
2	128.	Distinguish between constant current coulometry and controlled potential coulometry	2
2	129.	Define the term concentration polarisation	2
2	130.	Why there is a need for reference electrodes in potentiometric titration?	2
2	131.	What are electrochemical cells? With a schematic diagram differentiate between galvanic and electrolytic cells.	2
2	132.	What is the difference between primary coulometric secondary coulometric titrations?	2
2	133.	Write the classification of voltammetric techniques.	2
2	134.	Mention any two normal salt bridges which are used in routine analysis	2
2	135.	Differentiate electrolytic cell from a galvanic cell	2
2	136.	Give one example for each reference electrode, indicator/working electrode, an auxiliary electrode in voltammetry	2

2	137.	Calculate the time required for a constant current of 0.96 A to deposit 0.5 g of Co (II) as elemental cobalt on the surface of a cathode. The relative molar mass of cobalt=58.93	2
2	138.	A solution containing 0.25 g of copper as Cu^{2+} requires 20 minutes for complete deposition of copper at 1.25 A. Calculate the coulombs required and efficiency of the process (M_r=63.54).	2
2	139.	A 0.180 g of a purified organic acid sample was titrated coulometrically with hydroxide ions produced in 5 minutes by a constant current of 0.514 amps. Calculate the molar mass of the acid if n is one.	2
2	140.	How many minutes will it take for a current of 0.50 amps to cause the deposition of 0.50 g silver from a solution based on 80.0% current efficiency?	2
3	141.	With a schematic diagram explain the principle and working of TGA.	5
3	142.	With block diagram describe the principle and working of TGA.	5
3	143.	Explain the principle and the instrumentation of a modern thermobalance with a neat diagram.	5
3	144.	With block diagram describe the principle and working of a modern thermobalance.	5
3	145.	Describe the principle and working of TGA with a neat schematic diagram.	5
3	146.	Give the principle and instrumentation of a modern thermobalance with a neat diagram.	5
3	147.	Write the principle of TGA and explain the automatic thermogravimetric analysis.	5
3	148.	Give the principles of TGA. Write a note on automatic thermogravimetric analysis.	5
3	149.	Write the principle of TGA. Describe automatic thermogravimetric analysis.	5
3	150.	Write a note on the factors affecting the results of the TGA curve.	5
3	151.	Explain the factors affecting the results of the TGA curve.	5
3	152.	Describe the factors affecting the TGA curve.	5
3	153.	Write the factors affecting the TGA curve.	5
3	154.	Explain the factors affecting the TGA curve and give the applications of TGA.	5
3	155.	Describe the factors affecting the TGA curve. Write the applications of TGA.	5
3	156.	Write the factors affecting the TGA curve and give the applications of TGA.	5
3	157.	Write the thermogram where $CaC_2O_4.H_2O$ remains stable up to 100^0 C and $MgC_2O_4.H_2O$ up to 142^0 C. similarly CaC_2O_4 and MgC_2O_4 remain stable from 226^0 C to 398^0 C and 233^0 C to 397^0 C respectively. $CaCO_3$ is stable from 420-660^0 C. MgO and CaO are stable above 480^0 C and 842^0 C respectively. It is clear from the data that at 500^0 C, $CaCO_3$ and MgO are stable and at 900^0 C both metals exist as simple oxide.	5

3	158.	Write a short note on the applications of TGA.	5
3	159.	Discuss the applications of TGA.	5
3	160.	Elaborate on the applications of TGA.	5
3	161.	Outline the applications of TGA.	5
3	162.	Write the applications of TGA.	5
3	163.	Explain the applications of TGA.	5
3	164.	Describe the applications of TGA.	5
3	165.	Give the applications of TGA.	5
3	166.	Explain the theory and variables affecting the DTA curves.	5
3	167.	Describe the theory and variables affecting the DTA curves.	5
3	168.	Write a note on theory and variables affecting the DTA curves.	5
3	169.	Elaborate on the theory and variables affecting the DTA curves.	5
3	170.	Explain the principles and instrumentation of DTA with a neat diagram.	5
3	171.	With block diagram describe the principles and instrumentation of DTA.	5
3	172.	Describe the principles and instrumentationDTA with a neat diagram.	5
3	173.	Elaborate the principles and instrumentation of DTA with a neat diagram.	5
3	174.	Write a note on principle and instrumentation DTA with a neat diagram.	5
3	175.	Give the applications of differential thermal analysis.	5
3	176.	Discuss the applications of differential thermal analysis.	5
3	177.	Elaborate on the applications of DTA.	5
3	178.	Outline the applications of differential thermal analysis.	5
3	179.	Write the applications of DTA.	5
3	180.	Describe the applications of DTA.	5
3	181.	Describe the instrumentationDTA with a neat diagram. Differentiate TGA and DTA.	5
3	182.	Write a note on instrumentation DTA with a neat diagram. Give the differences between TGA and DTA.	5
3	183.	Explain the instrumentationDTA with an aschematic diagram. Write the differences between TGA and DTA.	5
3	184.	Outline the instrumentation DTA with a block diagram. Give the differences between TGA and DTA.	5
3	185.	Give the basic principle of DSC. Write the difference between DTA and DSC.	5
3	186.	Write the basic principle of DSC. Differentiate DTA and DSC.	5
3	187.	Give the principle of DSC. Differentiate between differential thermal analysis and differential scanning calorimetry.	5
3	188.	A neat and labeled diagram explains the power compensated and heat flux DSC.	5
3	189.	Explain the power compensated and heat flux DSC with a schematic	5

		diagram.	
3	190.	Outline the power compensated and heat flux DSC with a neat and labeled diagram	5
3	191.	Describe the power compensated and heat flux DSC with a block diagram.	5
3	192.	Write the power compensated and heat flux DSC with a schematic diagram.	5
3	193.	Write a note on the power compensated and heat flux DSC with a neat and labeled diagram	5
3	194.	Elaborate the power compensated and heat flux DSC with a block diagram.	5
3	195.	Write a note on thermomechanical and dynamic mechanical analysis. Write the applications of DSC.	5
3	196.	Explain thermomechanical and dynamic mechanical analysis. Give the applications of DSC.	5
3	197.	Describe thermomechanical and dynamic mechanical analysis. Write the applications of DSC.	5
3	198.	Discuss the applications of DSC.	5
3	199.	Elaborate on the applications of DSC.	5
3	200.	Outline the applications of DSC.	5
3	201.	Write the applications of DSC.	5
3	202.	Write a note on thermometric titrations highlighting the apparatus and its applications.	5
3	203.	Explain thermometric titrations highlighting the instrumentation and their applications.	5
3	204.	Describe the thermometric titrations highlighting the instrumentation and their applications with a block diagram.	5
3	205.	Outline the enthalpimetric titrations highlighting the instrumentation and their applications with a schematic diagram.	5
3	206.	Elaborate the enthalpimetric titrations highlighting the instrumentation and their applications with a neat labeled diagram.	5
3	207.	Write the theory and principles of thermometric titrations.	5
3	208.	Explain the theory and principles of thermometric titrations.	5
3	209.	Describe the theory and principles of thermometric titrations.	5
3	210.	Write a note on direct injection enthalpimetry.	5
3	211.	Explain direct injection enthalpimetry.	5
3	212.	Describe direct injection enthalpimetry.	5
1	213.	Compare atomic absorption and Flame emission instrument as to a) excitation source b) Sample cell and c) parameter which is measured for quantitative measurements.	5
1	214.	What are the advantages of the atomic absorption of a heated graphite atomizer over a flame atomizer?	5
1	215.	What are the advantages of total consumption as compared to a premix	5

		burner for atomic absorption use?	
1	216.	How are chemical interferences prevented or corrected for in atomic absorption determination?	5
1	217.	Explain the principle of flame emission and atomic absorption spectroscopy?	5
1	218.	Compare AAS and AES with respect to principle, advantages, and disadvantages	5
1	219.	Write the energy level diagram for a typical photoluminescent system and describe how an excited molecule can return to its ground state by a combination of several mechanistic steps.	5
1	220.	Give an account of interferences in Atomic absorption and Flame emission spectroscopy	5
1	221.	Discuss any five variables affecting luminescence spectroscopy	5
1	222.	Write the schematic diagram of typical premix and total consumption burners and explain their working. List out the advantages and disadvantages	5
1	223.	Write the sources of interferences in AAS. Discuss the steps in overcoming these interferences	5
1	224.	Write the block diagram of a typical AAS and FES and indicate the function of each component	5
1	225.	Explain the principle, instrumentation, and application of fluorimetry	5
1	226.	Explain the calibration curve procedure and standard addition technique for the quantitative analysis in FES	5
1	227.	Mention the different types of flames used in AAS. Discuss the effects of flame temperatures on atomic and emission spectra	5
1	228.	Discuss the principle, working, and important application of flame emission spectroscopy	5
1	229.	Compare premix burner and total consumption burner with respect to working, advantages, and disadvantages	5
1	230.	Brief the effect of concentration on fluorescence intensity. Substituents often have a marked effect on the luminescence of molecules, justify the statement	5
1	231.	Explain the phenomenon of the molecular photoluminescent process with an energy level diagram.	5
1	232.	Discuss the interferences in AAS and its methods of rectification briefly	5
1	233.	Compare the sensitivity and selectivity of molecular absorption measurements and luminescence measurements.	5
1	234.	Compare premix burner and total consumption burner. Which is more preferred and why?	5
1	235.	Explain with example the various factors which promote fluorescence	5
1	236.	Discuss the general principle of flame AAS and the process that occurs during flame atomization.	5
1	237.	Briefly discuss the principles of turbidimetry and Nephelometry with diagrams. Discuss the application of turbidimetry in the determination of sulfate in water	5

1	238.	Distinguish fluorometer from spectrofluorometer. Draw the block diagram of the spectrofluorometer and discuss its components	5
1	239.	Explain different types of interferences encountered in atomic absorption methods	5
1	240.	Discuss the application of atomic spectroscopy for quantitative analysis by calibration curve method	5
1	241.	What are flame atomizers? Discuss their importance of them in AAS	5
1	242.	How atomic absorption spectroscopy is different from atomic emission spectroscopy in principle and instrumentation	5
1	243.	Why sharp line source is required in AAS measurements? Explain the mechanism of operation of a hollow cathode lamp as a source in AAS	5
1	244.	Describe the instrumentation of AAS. Write the differences between AAS and FES. Mention the advantage of AAS and FES.	5
1	245.	Write the energy level diagram for a hypothetical molecular species showing luminescence, explain the processes: vibrational relaxation and internal conversion	5
1	246.	Describe the different processes that occur at the flame when a solution is aspirated with suitable examples.	5
1	247.	Write the principle, methodology, and working procedures of Nephelometry with any one example.	5
1	248.	Describe the components of fluorimetry. Why the most fluorescence instruments are double beam in design?	5
1	249.	Explain in detail the detection limit and sensitivity of Atomic absorption spectroscopy	5
1	250.	What are the sources of chemical interferences in AAS? Discuss briefly the ways of overcoming these interferences.	5
1	251.	Calculate the concentration of sulphate in the unknown sample if it was analyzed turbidimetrically as sulphate by addition of the barium chloride to varying concentrations of sulphate. The reading were as follows [SO_4^{2-}], ppm (%T): 15 (83.8), 26(69.2), 38.7 (49.0), 55.4 (28.3), unknown (37.2)	5
1	252.	Find out the concentration of silver from an unknown sample by Nephelometric analysis if precipitation was carried out with common salt. The reading observed were as follows: [Ag], ppm (%T): 1 (16), 3(37.3), 5 (51.5), 7 (63.8), 9(79.7) unknown (56.8)	5
1	253.	Calculate the concentration of lead from the unknown sample of lead chromate in Nephelometry by calibration curve which was constructed in a total volume of 50 ml. The reading were Volume of solution (absorbance): 2 (0.150), 4 (0.250), 6 (0.320), 8 (0.390) and 10 (0.480) unknown (0.350)	5
1	254.	In AAS measurements with cold vapour technique for Mercury from wastewater at 253.7 nm (with air-acetylene flame) gave the following reading of concentration of mercury (absorbance) shown in brackets 0(0), 2(0.053), 3.8(0.140), 5.8(0.160), 8(0.220), 9.6(0.260), 11.2(0.310), unknown (0.170)? Calculate the concentration of mercury in the unknown sample of water?	5
1	255.	The sample of water containing calcium was analyzed by atomic absorption technique at 422 nm with nitrous oxide acetylene flame when	5

		following readings for absorbance were observed: Concentration of calcium in ppm (absorbance) at 422 nm was 0(0), 1(0.090), 2.2(0.194), 3.10(0.273), 3.9(0.340), 5(0.440), 6(0.0525). The unknown solution had an absorbance of 0.380. What is the concentration of calcium in the unknown sample?	
1	256.	Calculate the concentration of Rubidium from the unknown sample when 1.2456g of salt was diluted to 100 ml and analyzed using the atomic absorbance technique at 780 nm. The absorbance reading against amount of Rubidium were mg/L (absorbance) were: 0.5 (0.240), 1.0(0.490), 2(1.030), 2.5(1.200), 4.0(1.900), unknown (1.210)	5
1	257.	What is the concentration of Zinc in the unknown sample when 2.979 g of Zinc was dissolved in dilute acid and made up to 100 ml and analyzed by AAS? The reading were Zn, μg/ml (absorbance): 1.0 (0.110), 3(0.300), 5(0.540), 6(0.670), 20(1.130), unknown sample (0.370)?	5
2	258.	Sketch and label a metal ion-selective electrode and explain the theory behind the indicator electrode of the first kind	5
2	259.	Describe the hydrogen electrode? What are its advantages and disadvantages?	5
2	260.	The response associated with the glass electrode is related to the pH of its environment by the expression, $E = k + 0.0591$ pH	5
2	261.	What is meant by ion-selective electrode? Explain with an example how the potential of electrodes varies with the concentration of ions.	5
2	262.	Explain the construction and working of silver-silver chloride and saturated calomel electrode	5
2	263.	Discuss the application of coulometric methods of analysis by controlled potential and controlled current methods	5
2	264.	Discuss the separation of metals by constant current electrogravimetry. Explain how the physical characteristics of deposited metals can be improved by the use of a depolarizer	5
2	265.	Explain the polarographic procedure in the quantitative analysis of metals.	5
2	266.	What are reference and indicator electrodes? Discuss the preparation and Nerstian mechanism of saturated calomel and silver-silver chloride electrode	5
2	267.	Explain the mechanism of pH determination by ion-selective electrodes. What precautions are to be taken during its measurement	5
2	268.	Explain the terms IR drop, concentration polarization, and kinetic polarisation. What precautionary steps are to be taken for controlling each of the polarisations?	5
2	269.	Explain how current is measured by constant current coulometry. How current efficiency can be achieved by the use of a mediator?	5
2	270.	Explain how voltammetry can be explored in the evaluation of the standard reduction potential of a reversible reaction	5
2	271.	Describe the construction of the glass membrane electrode and discuss the mechanism of its response for pH measurements	5
2	272.	Make a clear distinction between i) reduction and reducing agent ii) a galvanic cell and electrolytic cell iii) anode and cathode of an electrochemical cell iv) reversible electrochemical cell and irreversible	5

		electrochemical cell v) standard electrode potential and formal potential	
2	273.	What is voltammetry? How the voltammetry differs from electrogravimetry and coulometry? Explain the steps involved in stripping voltammetry	5
2	274.	How does a gas-sensing probe differ from other membrane electrodes? With an example, give the mechanism of the response of the gas sensing probe.	5
2	275.	Write short notes on i) Coulometric titrations ii) qualitative and quantitative analysis by voltammetry	5
2	276.	Describe the principle, construction, and working of enzyme electrodes for the determination of urea in body fluids.	5
2	277.	Write the principle of electrogravimetry. Explain its application	5
2	278.	Explain the following i) Reference electrode ii) working electrode iii) counter electrode iv) salt bridge v) liquid junction potential	5
2	279.	Explain the working, principle, and apparatus in the standard hydrogen electrode	5
2	280.	Describe the principle behind cyclic voltammetry with suitable illustration	5
2	281.	Write a note on coulometric and voltammetric methods of analysis	5
2	282.	What is the main classification of metallic indicator electrodes? Discuss each with a suitable example	5
2	283.	List out two reference electrodes and their properties with a neat labeled diagram and discuss electrochemical reactions for the development of potential at the electrode	5
2	284.	Explain liquid-based ion-selective electrodes and gas sensing electrodes	5
2	285.	What are reference electrodes? Discuss the common reference electrode used in potentiometry with half cell representation and half cell reaction along with net reaction	5
2	286.	Discuss coulometric titrations with respect to principle, apparatus, and its applications	5
2	287.	Draw typical cyclic voltammetry for a reversible redox system and explain its salient features of it with an example.	5
2	288.	Discuss in detail the classification of electrochemical methods for the analysis of electroactive species	5
2	289.	Write the difference between reference and indicator electrodes in potentiometric analysis. Describe the working principle of metallic indicator electrodes	5
2	290.	What is a crystalline membrane electrode? Discuss the construction and working principle of crystalline membrane ion-selective electrode to analyze fluoride, silver, and cadmium ions	5
2	291.	What is a reference electrode in potentiometry? Outline the construction and principle of working SCE and silver/silver chloride as reference electrodes	5
2	292.	Write the difference between controlled current coulometry and potentiostatic coulometry. Mention the advantages of coulometric titration over the conventional volumetric titrations	5
2	293.	Write the principle of electrogravimetry. Explain its application for the separation and quantification of copper, lead, and nickel in the alloy	5

2	294.	Outline the principle and construction of liquid-based ion-selective electrodes. How do these electrodes work when immersed in the test solution?	5
2	295.	What is an ideal reference electrode? Describe the construction and working of the Ag/AgCl electrode. Mention any two uses of this electrode	5
2	296.	Write the principle, methodology, and procedure involved in stripping voltammetry. Discuss any two applications in analytical chemistry	5
2	297.	The amount of iron in a meteorite was determined by a redox titration using potassium permanganate as the titrant. A 0.4185 g sample was dissolved in acid and the liberated Fe^{3+} quantitatively reduced to Fe^{2+}, using a reduction column. Titrating with 0.025 M potassium permanganate requires 41.27 ml to reach the endpoint. Determine the % Fe_2O_3 in the sample meteorite.	5
2	298.	The purity of a sample of $Na_2S_2O_3$ was determined by coulometric redox titration using I^- as a mediator, and I_3^- as the titrant. A sample weighing 0.1342 g is transferred to a 100 ml volumetric flask and diluted to volume with distilled water. A 10.0 ml portion is transferred to an electrochemical cell along with 25 ml of 1 M KI, 75 ml of a pH 7.0 phosphate buffer, and several drops of a starch indicator solution. Electrolysis at a constant current of 36.45 mA required 221.8 s to reach the starch indicator endpoint. Determine the purity of the sample.	5
2	299.	A 0.3619 g sample of tetrachloro-picolinic acid, $C_6HNO_2Cl_4$, is dissolved in distilled water, transferred to 1000 ml volumetric flask, and diluted volume. An exhaustive controlled potential electrolysis of 10.0 ml of the portion of this solution at spongy silver cathode requires 5.374 C of charge. What is the value of n for this reduction reaction? Write the reaction involved	5
2	300.	The purity of the sample of picric acid, $C_6H_3N_3O_7$ is determined by controlled potential coulometry converting the picric acid to triaminophenol , $C_6H_9N_3O$. A 0.2917 g sample of picric acid is placed in a 1000 ml volumetric flask and diluted to volume. A 10.0 ml portion of this solution is transferred to a coulometric cell and diluted till the platinum cathode is immersed. The exhaustive electrolysis of the sample requires 21.67 C of charge. Report the purity of the picric acid.	5
2	301.	The amount of iron in a 0.4891 g of ore was determined by a redox titration with $K_2Cr_2O_7$. The sample was dissolved in HCl and the iron was brought into the +2 oxidation state using jones's reductor. Titration to the diphenylamine sulphonic acid endpoint requires 36.92 mL of 0.02153 M $K_2Cr_2O_7$. Report the iron content of the ore as %w/w Fe_2O_3.	5
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Question Paper Pattern- Model Question Paper

Q.P Code: 85321

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE			
III SEMESTER –COURSE M.SC FINAL EXAMINATION			
SUBJECT : CHEMISTRY			
TITLE: INSTRUMENTAL METHODS OF ANALYSIS (HC)			
TIME: 3 HRS			**MAX MARKS: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	List out the various techniques involved in thermal methods of analysis	2
	b	Give the applications of TGA.	2
	c	List out the applications of differential thermal analysis.	2
	d	Give the principle and theory of thermometric titration.	2
	e	Define detection limit and sensitivity of an atomic absorption spectroscopy	2
	f	Define singlet and triplet states	2
	g	Why is the atomization process carried out in AAS? How the AAS differs from molecular absorption spectrometry?	2
	h	Why the most fluorescence instruments are double beam in design?	2
	i	Explain the terms a) charging current and b) migration current	2
	j	Define half wave potential, diffusion current, depolarizer DME, residual current	2
	k	What are the advantages and disadvantages of amperometric titrations over	2
	l	Discuss the pH limitations of ion selective electrodes	2
PART-B			
		Answer any Five of the following:	5 x 10= 50
2	a	Explain the principle and the instrumentation of a modern thermobalance with a neat diagram	5
	b	Write the thermogram where $CaC_2O_4.H_2O$ remains stable upto 100^0 C and $MgC_2O_4.H_2O$ upto 142^0 C. similarly CaC_2O_4 and MgC_2O_4 remains stable from 226^0 C to 398^0 C and 233^0 C to 397^0 C respectively. $CaCO_3$ is stable from 420-660^0 C. MgO and CaO are stable above 480^0 C and 842^0 C respectively. It is clear from the data that at 500^0 C, $CaCO_3$ and MgO are stable and at 900^0 C both metals exist as the simple oxide.	5
3	a	With block diagram describe the principles and instrumentation of DTA.	5
	b	Give the basic principle of DSC. Write the difference between DTA and DSC.	5
4	a	What are the advantages of a total consumption as compared to a premix burner for atomic absorption use?	5

	b	Compare AAS and AES with respect to principle, advantages and disadvantages	5
5	a	Explain the calibration curve procedure and standard addition technique for the quantitative analysis in FES	5
	b	Explain the phenomenon of molecular photoluminiscent process with an energy level diagram.	5
6	a	Sketch and label a metal ion selective electrode and explain the theory behind the indicator electrode of first kind	5
	b	The response associated with the glass electrode is related to the pH of its environment by the expression, $E = k + 0.0591\ pH$	5
7	a	What are reference and indicator electrodes? Discuss the preparation and Nerstian mechanism of saturated calomel and silver-silver chloride electrode	5
	b	The amount of iron in a meteorite was determined by a redox titration using potassium permanganate as the titrant. A 0.4185 g sample was dissolved in acid and the liberated Fe^{3+} quantitavely reduced to Fe^{2+}, using a reductor column. Titrating with 0.025 M potassium permanganate requires 41.27 ml to reach the end point. Determine the % Fe_2O_3 in the sample meteorite.	5
8	a	Write a note on direct injection enthalpimetry.	5
	b	The sample of water containing calcium was analysed by atomic absorption technique at 422 nm with nitrous oxide acetylene flame when following readings for absorbance were observed: Concentration of calcium in ppm (absorbance) at 422 nm was 0(0), 1(0.090), 2.2(0.194), 3.10(0.273), 3.9(0.340), 5(0.440), 6(0.0525). The unknown solution had an absorbance of 0.380. Calculate the concentration of calcium in the unknown sample by regression method?	5